Bridging the Regret Gap in Combinatorial Thompson Sampling: Worst-Case Guarantees and Algorithmic Refinement

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***Abstract*—This paper revisits *combinatorial Thompson sampling (CTS)* in the semi-bandit setting with sleeping arms, where only a subset of arms is available each round under combinatorial constraints. Such settings frequently arise in networking applications and pose significant challenges due to the need for combinatorial optimization over a dynamically changing action space. A canonical example is wireless mesh routing, where fluctuating link availability and routing constraints lead to a time-varying set of feasible paths between source and destination nodes.**

The existing works have three key limitations: (1) the lack of worst-case regret guarantees for CTS in semi-bandit settings, even without accounting for sleeping arms; (2) the absence of theoretical guarantees for CTS under adversarially varying arm availability; and (3) the subpar empirical performance of *CTS with Gaussian priors (CTS-G)*.

To address these issues, we propose CL-SG, a simple yet effective CTS-G variant that samples a single shared Gaussian seed each round to coordinate exploration across arms. Theoretically, CL-SG achieves a worst-case regret bound of $\tilde{O}(\sqrt{mNT})$ and a matching lower bound of $\Omega(\sqrt{mNT})$. Our analysis can also easily recover a regret bound for CTS-G. With experiments driven by real-world datasets, CL-SG consistently outperforms diverse benchmark algorithms such as CTS-G and CTS-B. We open-source our implementation and experiments to support reproducibility and further research.

I. Introduction

In this paper, we revisit *combinatorial Thompson sampling (CTS)* in the *stochastic semi-bandits with adversarial sleeping arms (sleeping semi-bandits)*, where an agent selects a subset of up to m arms from a dynamically changing set of N base arms over time, subject to predefined combinatorial constraints. Each arm is associated with a fixed but unknown reward distribution. In each round, upon playing a super arm (i.e., a subset of the available arms), the agent obtains a reward that is the sum of the individual rewards of the played arms drawn independently from their reward distributions, while also observing the individual reward associated with each played arm (i.e., semi-bandit feedback). The agent aims to accumulate as much reward as possible over T rounds.

Sleeping semi-bandits offer a natural framework for modeling networking scenarios with dynamic resource availability and partial feedback [1–5]. In routing, available links may change over time due to congestion, failures, or maintenance, and the algorithm must select feasible paths based on the current topology. Only the performance of selected links (e.g., delay, throughput) is observed. In wireless scheduling, user or channel availability may fluctuate due to fading or interference, and the scheduler selects a subset of users under interference or hardware constraints, observing feedback only for those scheduled. Sleeping semi-bandits are also a base scenario for more complicated settings, such as fairness constraints that are widely considered in networking domains [6–9].

Since the agent lacks prior knowledge of arm availabilities and reward distributions, it must rely on an online learning algorithm to make effective decisions. The performance of such algorithms is measured by *regret*, defined as the cumulative gap between the rewards of an oracle that selects the best available arms in each round and those obtained by the algorithm. Unlike the standard semi-bandit setting where the optimal action remains fixed, the optimal arms in the sleeping semi-bandit setting vary over time due to dynamic availability. This added non-stationarity increases the complexity of regret minimization and requires a more delicate balance between exploring the underlying reward distributions and exploiting the currently best-available arms.

The most promising strategies to balance exploration and exploitation are *upper confidence bound (UCB)* [10, 11] and *Thompson sampling (TS)*-based algorithms [12, 13], both of which are inspired by the principle of *optimism in the face of uncertainty (OFU)*. Both UCB and TS have been adapted to efficiently address the intricacies of sleeping semi-bandits.

UCB constructs upper confidence bounds around the empirical mean rewards to balance exploration and exploitation, and UCB maintains a deterministic approach to ensure that the agent remains optimistic about uncertain options. On the other hand, TS constructs data-dependent distributions (also known as posterior distributions) to model the true mean rewards of each arm. By sampling from posterior distributions, TS effectively balances the exploration-exploitation tradeoff through randomization, which has been confirmed, both empirically and theoretically, to be superior over UCB-based algorithms in many scenarios [14].

While theoretical guarantees of UCB-based algorithms for sleeping semi-bandits have been well established for both problem-dependent and worst-case (problem-independent) regret bounds[1], significant gaps remain in the literature regarding worst-case bounds for TS-based algorithms in this setting. Such bounds are crucial not only in practical scenarios where the reward gap Δ is unknown or ill-defined, particularly under sleeping constraints with a time-varying action space, but also as a theoretical benchmark for evaluating worst-case performance and enabling fair comparisons across algorithms.

Specifically, there are three limitations in the existing CTS algorithms for sleeping semi-bandits:

1) **Lack of worst-case guarantees.** While current research has established problem-dependent regret bounds for CTS comparable to those of UCB, scaling as $O(\ln T/\Delta)$, in the semi-bandit setting [15–18], no reasonable worst-case regret bound is known, let alone the sleeping semi-bandits. This is because such bounds are typically derived by tuning problem-dependent results, but existing CTS bounds suffer from unfavorable dependencies: for *CTS with Beta priors (CTS-B)*, the constant terms scale exponentially in m or N [15–17], and this exponential dependence has been proven unavoidable [17]. Even for *CTS with Gaussian priors (CTS-G)*, the best-known bound still involves high-degree polynomial dependencies on m and N [18]. As a result, there remains a significant gap in the frequentist, worst-case analysis of CTS for semi-bandits, although CTS-G appears more promising in this regard.
2) **Lack of theoretical guarantees under adversarial arm availability.** While theoretical guarantees exist for TS in settings with stochastic arm availability [2], no regret bounds are currently known for TS under adversarial arm availability, which is more appropriate for many real-world systems with unpredictable dynamics.
3) **Poor empirical performance.** Despite its more favorable theoretical structure, CTS-G consistently underperforms both UCB-based algorithms and CTS-B in practical evaluations, suggesting that its high variance or exploration inefficiency may limit its utility in real-world scenarios.

Our contributions address the above limitations as follows:

- We propose *combinatorial learning with a single Gaussian seed (CL-SG)*, a simple and effective variant of CTS-G that samples a single shared Gaussian value per round to drive highly efficient exploration.
- Theoretically, we establish the first worst-case regret bounds for TS-based algorithms in semi-bandits with (adversarially) sleeping arms. Our algorithm, CL-SG, achieves a near-optimal upper bound of $\tilde{O}(\sqrt{mNT})$ and a matching lower bound of $\Omega(\sqrt{mNT})$. This also yields the first such guarantee for standard CTS-G, resolving a long-standing open question where previous CTS analyses were inherently problem-dependent.

[1]Problem-dependent bounds rely on instance-specific quantities such as the reward gap Δ between optimal and suboptimal actions, usually scaling as $O(\ln T/\Delta)$, whereas worst-case bounds hold uniformly across all instances, typically scaling as $O(\sqrt{T})$.

  Our core analysis techniques depart from traditional gap-based techniques and instead directly analyze a worst-case regime, combining a ghost-sample argument with a refined anti-concentration analysis for correlated Gaussian estimates (see Lemma 2 in Sec. V).
- Empirically, CL-SG consistently outperforms benchmark algorithms such as CTS-G and CTS-B in experiments grounded in real-world datasets. We also release our implementation and experimental framework as open source to support reproducibility and further research.

II. The Sleeping Semi-Bandit Problem

We consider a sleeping semi-bandit problem with a fixed set of N base arms denoted by $[N] := \{1, 2, \ldots, N\}$. Each base arm $a \in [N]$ is associated with a fixed but unknown reward distribution p_a supported on $[0, 1]$ with its mean denoted by r_a. Denote by $\Theta \subseteq 2^{[N]}$ the *feasible set* consisting of all possible *solutions* satisfying some certain constraints. Each feasible solution $A \in \Theta$ can be viewed as a super arm, which can be made up of more than one base arm. Let $m := \max_{A\in\Theta} |A|$ denote the maximum cardinality among all super arms, i.e., the maximum number of base arms in a super arm.

Different from the standard combinatorial bandits setting [19], where the learning agent faces up to a fixed decision set Θ in all the rounds, in the sleeping semi-bandit setting, in each round $t = 1, 2, \ldots, T$, a time-varying feasible set $\Theta_t \subseteq \Theta$ is revealed to the learning agent. The feasible set could be generated in an adversarial way. Then, the learning agent plays a super arm $A_t \in \Theta_t$, observes the random rewards $r_{a,t} \sim p_a$ for all the base arms $a \in A_t$, and obtains a reward $\sum_{a\in A_t} r_{a,t}$. The goal of the learning agent is to choose a sequence of super arms to play to accumulate as much reward as possible over a finite number of T rounds. Since Θ_t is revealed at the beginning of each round t, we let $A_t^* := \arg\max_{A\in\Theta_t} \sum_{a\in A} r_a$ denote the optimal super arm in round t. Then, the T-round (pseudo)-regret can be expressed as

$$\mathcal{R}(T) := \sum_{t=1}^{T} \mathbf{E}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} r_a\right], \tag{1}$$

where the expectation is taken over Θ_t and A_t. Note that A_t^* is also random, which is determined by Θ_t.

III. Related Works

Given the foundational importance and practical relevance of UCB and TS in stochastic bandits, our discussion will primarily focus on adapting these algorithms for stochastic semi-bandits and stochastic sleeping semi-bandits.

Semi-Bandits. Semi-bandits are a special case of combinatorial bandits [20, 21], where the reward of each played base arm can be observed. The performance of UCB-based algorithms for semi-bandits has been well studied. A sublinear problem-dependent regret upper bound is derived in [22] for a UCB-based algorithm called CombUCB. Later, the authors of [19] not only improved the problem-dependent regret bound to $O(mN\ln(T)/\Delta)$ but also derived an $O(\sqrt{mNT\ln T})$

worst-case regret bound for CombUCB. In [19] and [23], an $\Omega(\sqrt{mNT})$ minimax regret lower bound was derived for the combinatorial bandits. When the reward distributions are mutually independent, it is proved in [24] that the UCB-based algorithm can achieve a better problem-dependent regret bound of $O(N\ln^2(m)\ln(T)/\Delta)$, and a worst-case regret bound of $\sqrt{N\ln^2(m)T\ln T}$.

Regarding TS-based algorithms for semi-bandits, the authors of [15] proved the first $O\left(mN\ln(T)/\Delta\right)$ problem-dependent regret bounds of CTS-B. The idea of CTS-B is to use Beta distributions to model the mean reward of each arm's reward distribution. Then, the authors of [16] improved the results of [15] to $O\left(\frac{N(\ln m)^2}{\Delta}\ln T + \frac{Nm^3}{\Delta^2} + m\left(\frac{m^2+1}{\Delta}\right)^{2+4m}\right)$, when the arm distributions are mutually independent. However, both the problem-dependent bounds in [15] and [16] contain a term that exponentially increases with the size of the optimal solutions. Later, the authors of [17] proved that this exponential term is unavoidable for CTS-B. Subsequently, using Gaussian priors, a significant improvement is made in [18] by reducing this exponential dependency to a polynomial term: $O\left(\frac{N\ln m}{\Delta}\ln T + \frac{N^2 m\ln m}{\Delta}\ln\ln T + P\left(m, N, \frac{1}{\Delta}, \Delta\right)\right)$. However, the polynomial term $P\left(m, N, \frac{1}{\Delta}, \Delta\right)$ still has a degree approximately 30 in m and 10 in N. Thus, it is still difficult to obtain reasonable worst-case regret bounds from tuning the problem-dependent bounds [25].

On the other hand, the authors of [26] gave an $O\left(\max\left\{N\sqrt{T\ln T}, N^2\right\}\right)$ worst-case Bayesian regret bound of CTS-B.

Thus, while problem-dependent bounds of TS-based algorithms are well-studied, non-Bayesian worst-case bounds of TS-based algorithms for semi-bandits have remained an open challenge for a long time.

Sleeping Semi-Bandits. All the aforementioned works assume that the arm set from which the learning agent can play is fixed over all T rounds, i.e., all the arms are always available and ready to be played. However, in practice, some of the arms may not be available in some rounds. Therefore, a bunch of literature studied the setting of sleeping semi-bandits [1–7]. In sleeping bandits, the set of available arms for each round, i.e., the availability set, can vary. For the simplest version of sleeping bandits, the problem-dependent regret bounds of UCB-based algorithms and TS-based algorithms have been analyzed in [1] and [2], respectively. Regarding the sleeping semi-bandits, the authors of [3] proposed a UCB-based algorithm and derived a problem-dependent regret bound. The authors of [7] studied a variant of sleeping semi-bandits with fairness constraints, and if relaxing the fairness constraints, they gave a worst-case bound of $O(\sqrt{mNT\ln T})$ for UCB-based algorithms. Both the above works assume a stochastic availability set. The same-order worst-case upper regret bound for UCB on a non-stochastic (adversarial) availability set was also obtained in [27].

However, no prior work has established a **worst-case, non-Bayesian regret bound** for CTS in (sleeping) semi-bandits with stochastic or **adversarial arm availability**. Since CTS-G has demonstrated more favorable scaling in existing problem-dependent bounds [18], raising the question of whether a reasonable worst-case guarantee is achievable. This motivates our study of CL-SG, where the regret analysis of CL-SG can be easily extended to CTS-G and prove that a reasonable worst-case regret bound does, in fact, exist.

IV. The CL-SG Algorithm

In this section, we bridge the aforementioned gaps by introducing *Combinatorial Learning with a Single Gaussian Seed (CL-SG)*, presented in Alg. 1.

To formalize the setup, let $n_{a,t} := \sum_{\tau=1}^{t-1}\mathbf{1}[a \in A_\tau]$ denote the total number of times that base arm $a \in [N]$ has been pulled at the beginning of round t. Let $\hat{r}_{a,n_{a,t}} := \frac{\sum_{\tau=1}^{t-1}\mathbf{1}[a\in A_\tau]\cdot r_{a,\tau}}{n_{a,t}}$ denote the empirical mean of base arm a at the beginning of round t, which is the average of $n_{a,t}$ i.i.d. random rewards $r_{a,\tau}$ drawn from reward distribution p_a.

While the idea of using shared Gaussian noise has appeared in other contexts such as single-arm decision-making in reinforcement learning [28], CL-SG introduces this technique in the combinatorial semi-bandit setting with sleeping arms. This setting presents additional challenges due to its combinatorial action space and dynamic arm availability. CL-SG replaces CTS-G's per-arm Gaussian sampling with a single shared Gaussian sample per round, resulting in reduced variance and improved empirical performance.

Specifically, while the standard CTS-G algorithm independently samples from a Gaussian posterior $\mathcal{N}(\hat{r}_{a,n_{a,t}}, \frac{\gamma m\ln t}{n_{a,t}+1})$ for each available arm a as mean reward estimator, CL-SG draws a single shared Gaussian sample $w_t \sim \mathcal{N}(0,1)$ per round. It then constructs the reward estimate for each arm as $\bar{r}_{a,t} = \hat{r}_{a,n_{a,t}} + w_t \cdot \sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}$ where $\gamma > 0$ is an exploration parameter. Thus, each $\bar{r}_{a,t}$ still marginally follows a Gaussian distribution $\mathcal{N}(\hat{r}_{a,n_{a,t}}, \frac{\gamma\ln t}{n_{a,t}+1})$, but the estimates across arms are now perfectly correlated through the shared w_t, in contrast to CTS-G where all estimates are independent. The algorithm then selects the action $A_t = \arg\max_{A\in\Theta_t}\sum_{a\in A}\bar{r}_{a,t}$ from the feasible set Θ_t in round t.

Algorithm 1 The CL-SG Algorithm

Require: arm set $[N]$, exploration rate γ

Initialize $n_{a,1} = 0$ and $\hat{r}_{a,0} = 0$ for all base arms $a \in [N]$
for $t = 1, 2, \ldots$ **do**
    Observe feasible set Θ_t
    Draw $w_t \sim \mathcal{N}(0,1)$
    Construct $\bar{r}_{a,t} = \hat{r}_{a,n_{a,t}} + w_t \cdot \sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}$ for all base arms $a \in [N]$
    Play super arm $A_t = \underset{A\in\Theta_t}{\arg\max}\sum_{a\in A}\bar{r}_{a,t}$
    Observe $r_{a,t} \sim p_a$ for all base arms $a \in A_t$ and update $n_{a,t}$ and $\hat{r}_{a,n_{a,t}}$ for all $a \in A_t$.
end for

V. Analytical Results

Our main result in Theorem 1 builds on two key lemmas. Before stating them, we introduce the necessary notations. Let $\mathcal{E}_t := \left\{|r_a - \hat{r}_{a,n_{a,t}}| \le \sqrt{\frac{3\ln(Nt)}{n_{a,t}+1}}, \forall a \in [N]\right\}$ be the event that the empirical means are close to their true means at the beginning of round t. Let $\mathcal{F}_t$ denote the observed history up to round t, and define $\mathbf{E}_{\Theta_t}[\cdot] := \mathbf{E}[\cdot \mid \Theta_t]$.[2] We also set $t' = \max\{\sqrt{m}, 4\}$. With this setup, we now decompose the regret in (1) via the following lemma.

Lemma 1. *The regret of CL-SG can be decomposed as follows.*

$$\mathcal{R}(T) \le O(mt') + \underbrace{\sum_{t=t'}^{T} \mathbf{E}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} \bar{r}_{a,t}\right]}_{=:I_1,\ \textit{optimism term}}$$
$$+ \underbrace{\sum_{t=t'}^{T} \mathbf{E}\left[\mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t} (\bar{r}_{a,t} - r_a)\, \mathbf{1}[\mathcal{E}_t]\right]\right]}_{=:I_2,\ \textit{deviation term}}.$$

Proof of Lemma 1. By definition of regret in (1), we have that

$$\begin{aligned}\mathcal{R}(T) =& \sum_{t=1}^{t'-1} \mathbf{E}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} r_a\right] + \sum_{t=t'}^{T} \mathbf{E}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} r_a\right] \\ \le & mt' + \underbrace{\mathbf{E}\left[\sum_{t=t'}^{T}\left(\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} r_a\right)\mathbf{1}[\mathcal{E}_t]\right]}_{=:\text{(a)}} \\ &+ \underbrace{\mathbf{E}\left[\sum_{t=t'}^{T}\left(\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} r_a\right)\mathbf{1}[\overline{\mathcal{E}_t}]\right]}_{=:\text{(b)}}.\end{aligned} \tag{2}$$

where the inequality is due to the fact that $\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} r_a \le m$. (b) is bounded by $m\frac{\pi^2}{3N^2}$, as the union bound and Hoeffding's inequality ensures $\sum_{t=1}^{T} \Pr(\overline{\mathcal{E}_t}) \le \frac{\pi^2}{3N^2}$. Then, the lemma follows by noticing that (a) $\le I_1 + I_2$. $\square$

The deviation term I_2 is easy to analyze as we can observe A_t, and is upper bounded by $\tilde{O}(\sqrt{mNT})$ via using concentration bounds (See Lemma 4 in Appendix C). The central question is how to upper bound the optimism term I_1, which measures the gap between the *maximum amount of true reward* $\sum_{a\in A_t^*} r_a$ that the learning agent could achieve and the *expected maximum amount of reward* $\sum_{a\in A_t} \bar{r}_{a,t}$ that the learning agent can observe in round t. The following lemma is one of our key technical contributions that bounds I_1.

Lemma 2. *The optimism part in CL-SG satisfies that*

$$I_1 \le 8\sqrt{2\gamma}\,\Phi(-\sqrt{4/\gamma})^{-1} \ln T \sqrt{mNT}. \tag{3}$$

[2]We note that such a definition $\mathbf{E}_{\Theta_t}[\cdot]$ applies pointwise for any realized Θ_t, with no distributional assumption made on the availability process.

Proof Sketch of Lemma 2. Our core technique for bounding the gap between $\sum_{a\in A_t^*} r_a$ and $\sum_{a\in A_t} \bar{r}_{a,t}$ involves adapting a ghost-sample-based argument to the combinatorial semi-bandit setting with time-varying feasible sets. While related ideas have appeared in prior work [29], our analysis handles additional challenges introduced by combinatorial actions and sleeping arms. The proof consists of the following three steps; full details are deferred to Appendix B.

Step 1: Gap Conversion. We first relate the gap between the optimal reward and the estimated reward via an anti-concentration argument. Specifically, we convert the difference $\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} \bar{r}_{a,t}$ to the deviation of $\sum_{a\in A_t} \bar{r}_{a,t}$ from its expectation:

$$\mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t} \bar{r}_{a,t}\right] \le 2\Phi(-\sqrt{4/\gamma})^{-1} \cdot \mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t} \bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t} \bar{r}_{a,t}\right]\right)^+\right]. \tag{4}$$

This step relies on a reverse application of Markov's inequality, along with a careful characterization of the anti-concentration behavior of correlated Gaussian variables (see Lemma 5 in Appendix D).

Step 2: Ghost-Sample Analysis. To remove the activation function in the above bound, we introduce a ghost sample $\tilde{w}_t \sim \mathcal{N}(0,1)$ that is independent of w_t and define $\tilde{r}_{a,t} = \hat{r}_{a,t} + \tilde{w}_t \cdot \sqrt{\frac{\gamma \ln t}{n_{a,t}+1}}$ for all a. We then show that:

$$\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t} \bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t} \bar{r}_{a,t}\right]\right)^+\right] \le \mathbf{E}_{\Theta_t}\left[\left|\sum_{a\in A_t} \bar{r}_{a,t} - \sum_{a\in A_t} \tilde{r}_{a,t}\right|\right]. \tag{5}$$

Step 3: Aggregation Over Time. Finally, summing over T rounds and applying Hölder's inequality, we obtain:

$$\mathbf{E}\left[\sum_{t=1}^{T}\left|\sum_{a\in A_t} \bar{r}_{a,t} - \sum_{a\in A_t} \tilde{r}_{a,t}\right|\right] \le 4\ln T\sqrt{2\gamma mNT}. \tag{6}$$

Combining the above steps completes the proof. $\square$

We now derive the regret upper bound for CL-SG as follows.

Theorem 1. *(1) The regret of CL-SG is $O\left(\ln(T)\sqrt{mNT}\right)$. (2) There exists a problem instance such that CL-SG suffers $\Omega\left(\sqrt{mNT}\right)$ regret.*

a) Discussion: Theorem 1 shows that CL-SG achieves a worst-case regret bound of $\tilde{O}(\sqrt{mNT})$. Compared to UCB-based algorithms for sleeping semi-bandits, which achieve $O(\sqrt{mNT\ln T})$ regret under stochastic availability assumptions [3, 7], our bound includes an extra factor of $\sqrt{\ln T}$. However, a key distinction is that **our analysis does not require any stochastic assumptions on arm availability**, making the result applicable in more general settings.

Importantly, the regret analysis can be easily extended to CTS-G (only a slight modification on the anti-concentration bound in Step 1 of Lemma 2), making it the **first worst-case regret bound for a CTS algorithm in semi-bandits** that avoids exponential or high-degree polynomial dependence on m or N, **addressing a long-standing open question in the literature**. Furthermore, our lower bound for CL-SG matches the known minimax lower bound of $\Omega(\sqrt{mNT})$ for combinatorial bandits [23]. While our current upper bound still includes an extra $\ln T$ factor, this gap is relatively mild and our lower bound suggests that CL-SG may be further refined to achieve a fully minimax-optimal bound in future work.

b) Upper Bound Proof: Invoking Lemmas 1, 2, and 4, we can bound the regret as follows:

$$\mathcal{R}(T) \leq \left(4\sqrt{\gamma} + 8\sqrt{2\gamma}\Phi(-\sqrt{4/\gamma})^{-1}\right) \ln T\sqrt{mNT} + 2\sqrt{6mNT\ln T} + O(mt').$$

Since $\gamma > 0$ is a constant, we can numerically tune it to minimize the coefficient of the leading term. Through grid search over $\gamma \in [0.0001, 100]$, the optimal value is found to be $\gamma = 4.57$, yielding a minimized coefficient of approximately 144.43.

c) Lower Bound Proof Sketch: Inspired by Theorem 1.4 in [13], we construct a path selection problem with N links (base arms) and K paths (super arms) of m links, ensuring $N = mK$. This reduces the semi-bandits to the K independent path selection problem, as shown in Fig. 1.

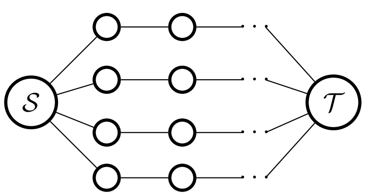


Fig. 1: Problem instance for the lower-bound proof. Nodes $\mathcal{S}$ and $\mathcal{T}$ are the starting and ending points for each path.

Without loss of generality, we assume path A_1 is the unique optimal super arm. Let $\Delta := \sqrt{K\ln(K)/T}$. We design that each base arm in A_1 always yields a reward of $\sqrt{\gamma}\Delta$ with probability 1, and each in suboptimal super arms always yields 0 with probability 1. The main idea of the lower bound is to show that the probability of playing a suboptimal super arm is always at least a constant, i.e. $\Pr\left(\exists A \in \Theta \setminus A_1 : A_t = A \mid \mathcal{F}_{t-1} = F_{t-1}\right) > c_0$ for some constant $c_0 > 0$. The main technical challenge arises from the fact that CL-SG uses a single shared Gaussian sample across all arms, which induces correlations among the reward estimates and complicates standard decoupling arguments.

To overcome this, we introduce a novel analysis centered on identifying a constant $\alpha \in (0, 1)$ such that the optimal and suboptimal super arms can be effectively "decoupled" after round αT. We then show that from round αT onward, any fixed suboptimal super arm incurs a constant per-round regret gap of at least $c_0\sqrt{\gamma}\Delta \cdot m$. This yields a total regret lower bound of $(1-\alpha)T \cdot c_0 \cdot \sqrt{\gamma}\Delta \cdot m = \Omega(\sqrt{mNT\ln(N/m)})$. More details can be found in Appendix A.

VI. Experiments

In this section, we evaluate our algorithm in the context of network routing with sleeping semi-bandits. Specifically, we assess the performance of CL-SG against the standard CTS-G algorithm, which independently samples for each arm according to $\mathcal{N}\left(\hat{r}_{a,n_{a,t}}, \frac{\gamma m \ln t}{n_{a,t}+1}\right)$, and we examine the effect of varying the exploration parameter γ. In addition, we compare against the following baselines, each selecting the action $A_t = \arg\max_{A\in\Theta_t} \sum_{a\in A} \theta_{a,t}$ with $\theta_{a,t}$ defined as follows:

- CTS-B [15]: $\theta_{a,t} \sim \text{Beta}(\hat{r}_{a,n_{a,t}} n_{a,t} + 1, n_{a,t} - \hat{r}_{a,n_{a,t}} n_{a,t} + 1)$.
- BG-CTS [18]: $\theta_{a,t} \sim N\left(\hat{r}_{a,n_{a,t}}, 2g(t)\sigma^2/n_{a,t}\right)$, with $\sigma^2 = 1/4$ for Bernoulli rewards and $g(t)$ defined as in [18].
- CombUCB [19]: $\theta_{a,t} = \hat{r}_{a,n_{a,t}} + \sqrt{1.5\ln t/n_{a,t}}$.

We evaluate these algorithms under two routing scenarios:

- Setting 1 (Synthetic Network): This is a controlled numerical experiment based on a wireless mesh network of 16 nodes in a 4×4 grid with a total of 24 links. Among the links, one predefined path of four hops yields Bernoulli rewards with mean 0.9 per link, while the remaining links yield Bernoulli rewards with mean 0.8. All links have a uniform availability probability of 0.75.
- Setting 2 (Real-World Network): This setting uses real-world traces from the UCSB MeshNet dataset [30, 31], which provides per-minute neighborhood tables. Each row records the *expected transmission time (ETT)* between a node and its neighbors. We define the reward for each link as $1 -$ normalized ETT. Since link availability varies over time, this naturally fits into the sleeping semi-bandit framework.

All reported results are averaged over 100 independent runs.

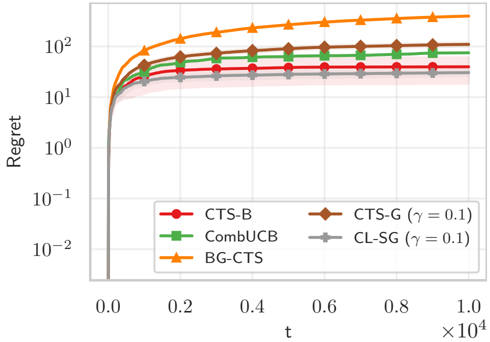
a) Comparison of the Regret: The regret results over $T = 10^4$ rounds are shown in Fig. 2 with shaded areas indicating 97.5% confidence intervals. The confidence intervals for some algorithms are not easily visible due to their small size.

In both settings, CL-SG draws a minimal number of Gaussian random samples in each round, enhancing its efficiency. This reduction in randomness improves robustness, even for a large γ, preventing excessive exploration. As a result, CL-SG achieves superior performance, surpassing CTS-B, BG-CTS, and CTS-B. This indicates the efficiency of CL-SG's design in optimizing the exploration-exploitation trade-off more effectively than its counterparts.

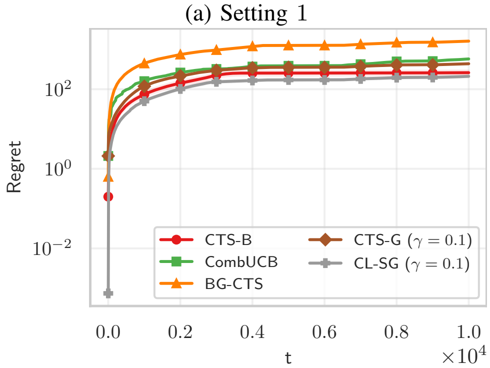
b) Effect of Different Exploration Rates: We also compare the performance of CTS-G and CL-SG under different exploration rates, i.e., $\gamma = 0.01, 0.1, 0.5$, and 1. The results are shown in Fig. 3. We can observe that in both settings, CL-SG and CTS-G achieve the lowest regret when $\gamma = 0.01$. If we continue to increase γ, both algorithms suffer a larger regret. This suggests that a certain lower level of randomness (exploration) is more effective in practice.

VII. Conclusion

This paper addresses a long-standing open problem by establishing the first worst-case regret bounds for TS-based algorithms in sleeping semi-bandit settings. We propose CL-SG, a variant of CTS-G that draws only a single shared

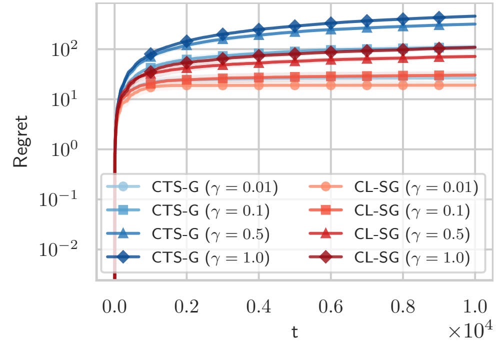


(a) Setting 1

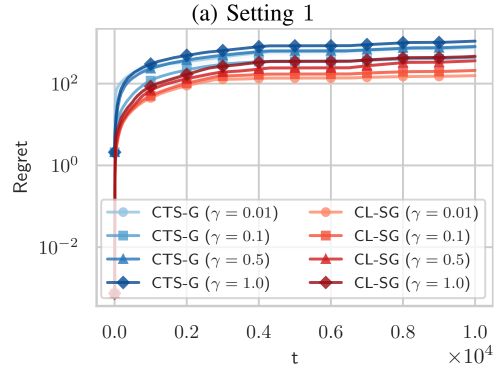


(b) Setting 2

Fig. 2: Comparison of the regret for both settings with $\gamma = 0.1$ for both CTS-G and CL-SG.



(a) Setting 1



(b) Setting 2

Fig. 3: Effect of different exploration rates $\gamma = 0.01, 0.1, 0.5,$ and 1 for CTS-G and CL-SG.

Gaussian sample per round. CL-SG achieves near-optimal performance, with a worst-case upper bound of $\tilde{O}(\sqrt{mNT})$ and a matching lower bound of $\Omega(\sqrt{mNT})$. Empirically, CL-SG significantly outperforms existing benchmarks such as CTS-B and CombUCB, demonstrating both improved accuracy and efficiency across diverse environments.

Looking ahead, we aim to theoretically characterize the optimal trade-off between randomness and exploration, and to further obtain the minimax-optimal regret bounds. The authors have provided public access to their code at **https://tinyurl.com/infocom26TS**.

Acknowledgment

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Appendix

Notations: Let $\mathcal{F}_{t-1}$ denote the history of past actions and rewards until the end of round $t-1$. Recall that $\mathbf{E}_{\Theta_t}[\cdot] := \mathbf{E}[\cdot \mid \Theta_t]$ and $\Pr_{\Theta_t}(\cdot) := \Pr(\cdot \mid \Theta_t)$, and $\mathcal{E}_t := \left\{\forall a \in [N] : |r_a - \hat{r}_{a,n_{a,t}}| \leq \sqrt{\frac{3 \ln Nt}{n_{a,t}+1}}\right\}$ is the high-probability event that the empirical mean is close to the true mean reward for arm a. Denote by $\overline{\mathcal{E}_t}$ the complementary event of $\mathcal{E}_t$. Recall that $\tilde{r}_{a,t} := \hat{r}_{a,n_{a,t}} + \tilde{w}_t \sqrt{\frac{\gamma \ln t}{n_{a,t}+1}}$, where $\tilde{w}_t \sim \mathcal{N}(0,1)$ is i.i.d. of w_t for CL-SG.

Fact 1. *For a Gaussian distributed random variable Z with mean μ and variance δ^2, for any z, we have that*

$$\frac{1}{4\sqrt{\pi}} \cdot e^{-7z^2/2} \leq \Pr(|Z - \mu| > z\sigma) \leq \frac{1}{2} e^{-z^2/2}, \tag{7}$$

and for any $z > 0$,

$$\Pr(Z - \mu > z\sigma) \geq \frac{1}{\sqrt{2\pi}} \frac{z}{z^2+1} e^{-\frac{z^2}{2}}. \tag{8}$$

Fact 2. *Let $X_1, \ldots, X_N$ be N real random variables with $X_i \sim \operatorname{subG}\left(\sigma^2\right), i = 1, \ldots, N$, not necessarily independent. Then,*

$$\mathbf{E}\left[\max_{i=1,\ldots,N} |X_i|\right] \leq \sigma\sqrt{2\ln(2N)}.$$

A. Proof of the Lower Bound of CL-SG

Lower Bound Proof in Theorem 1. The main challenge in the analysis stems from the shared Gaussian seed across all arms, which induces dependencies among paths and breaks their independence. We consider a path-selection problem with K paths and N = mK base arms, where each path consists of m disjoint links, as illustrated in Fig. 1. The availability set is fixed across all T rounds, $\Theta_t = \Theta := \{A_1, \ldots, A_K\}$, with A_1 being the unique optimal path. Each base arm follows a Bernoulli distribution. Let $\Delta := \sqrt{K/T}$. Base arms in A_1

always yield reward $\sqrt{\gamma}\Delta$, while all others yield zero. Since paths do not share base arms, the number of observations for any base arm in A equals $Q_A(t)$, the number of times path A is selected by round t. Let $c := 1/6$ and define the event $B_t^* := \{Q_{A_1}(t) > t - cT\}$, indicating that the optimal path has been sufficiently explored by round t. We lower bound the regret by considering two exhaustive cases depending on whether B_t^* occurs.

If B_t^* is not true for some round $t \in [T]$, we have the total number of times of playing sub-optimal super arms until the beginning of round t is $\sum_{A \in \Theta \setminus A_1} Q_A(t) = t - Q_{A_1}(t) \geq cT$. This lower bound implies the total regret from round 1 to the end of round $t-1$ is at least $cT \cdot m \cdot \sqrt{\gamma}\Delta = \Omega(Tm\sqrt{K/T}) = \Omega(Tm\sqrt{N/(mT)}) = \Omega(\sqrt{mNT})$. Note that this lower bound is also a regret lower bound for the total regret from round 1 to the end of round T.

If B_t^* is true for all rounds $t \in [T]$, let $\alpha := \frac{5}{6}$, and the total regret from round 1 to the end of round T is lower bounded by the total regret from round $t = \alpha T$ to the end of round T. In each round $t \geq \alpha T$, we have the following inequalities:

$$\sum_{A \in \Theta \setminus A_1} Q_A(t) = t - Q_{A_1}(t) \leq t - (t - c \cdot T) = c \cdot T, \quad (9)$$

$$Q_{A_1}(t) > t - c \cdot T \geq \alpha \cdot T - c \cdot T = (\alpha - c) \cdot T, \quad (10)$$

$$Q_A(t) \leq c \cdot T, \quad \forall A \in \Theta \setminus A_1. \quad (11)$$

From (10) and (11), for each sub-optimal super arm $A \in \Theta \setminus A_1$, we have

$$\sqrt{\frac{Q_A(t)+1}{Q_{A_1}(t)+1}} \leq \sqrt{\frac{cT+1}{(\alpha - c)T + 1}} = \sqrt{\frac{T/6+1}{4T/6+1}} \leq \frac{1}{2}, \quad (12)$$

which gives $1 - \sqrt{\frac{Q_A(t)+1}{Q_{A_1}(t)+1}} \geq \frac{1}{2}$.

Let $c_0 := \frac{1}{8\sqrt{\pi}} e^{-28/3}$. We show that for all $t \geq \alpha T$, conditioned on B_t^*, a sub-optimal super arm is played with probability at least c_0.

Since B_t^* is $\mathcal{F}_{t-1}$-measurable, for any realization F_{t-1} such that B_t^* holds, the probability of selecting a sub-optimal super arm satisfies

$$\begin{aligned}
&\Pr\left(\exists A \in \Theta \setminus A_1 : A_t = A \mid \mathcal{F}_{t-1} = F_{t-1}\right) \geq \Pr\left(\exists A \in \Theta \setminus A_1 : \sum_{a \in A} \bar{r}_{a,t} > \sum_{b \in A_1} \bar{r}_{b,t} \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&= \Pr\left(\exists A \in \Theta \setminus A_1 : \sum_{a \in A} \hat{r}_{a,n_{a,t}} + w_t\sqrt{\frac{\gamma \ln t}{n_{a,t}+1}} > \sum_{b \in A_1} \hat{r}_{b,n_{b,t}} + w_t\sqrt{\frac{\gamma \ln t}{n_{b,t}+1}} \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&\overset{(a)}{=} \Pr\left(\exists A \in \Theta \setminus A_1 : \sum_{a \in A} w_t\sqrt{\frac{\gamma \ln t}{n_{a,t}+1}} > \sum_{b \in A_1}\left(\sqrt{\gamma}\Delta + w_t\sqrt{\frac{\gamma \ln t}{n_{b,t}+1}}\right) \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&\overset{(b)}{=} \Pr\left(\exists A \in \Theta \setminus A_1 : \sum_{a \in A} w_t\sqrt{\frac{\gamma \ln t}{Q_A(t)+1}} > \sum_{b \in A_1}\left(\sqrt{\gamma}\Delta + w_t\sqrt{\frac{\gamma \ln t}{Q_{A_1}(t)+1}}\right) \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&= \Pr\left(\exists A \in \Theta \setminus A_1 : w_t\sqrt{\frac{\ln t}{Q_A(t)+1}} > \Delta + w_t\sqrt{\frac{\ln t}{Q_{A_1}(t)+1}} \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&\geq \Pr\left(\exists A \in \Theta \setminus A_1 : w_t\left(1 - \sqrt{\frac{Q_A(t)+1}{Q_{A_1}(t)+1}}\right) > \Delta\sqrt{Q_A(t)+1} \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&\overset{(c)}{\geq} \Pr\left(\exists A \in \Theta \setminus A_1 : w_t \cdot \frac{1}{2} > \Delta\sqrt{Q_A(t)+1} \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&= 1 - \underbrace{\Pr\left(w_t \leq 2\Delta\sqrt{Q_A(t)+1}, \forall A \in \Theta \setminus A_1 \mid \mathcal{F}_{t-1} = F_{t-1}\right)}_{\lambda},
\end{aligned} \quad (13)$$

where step (a) uses the fact that for any base arm in a sub-optimal super arm, the empirical mean is 0, whereas for any base arm in the optimal super arm, the empirical mean is $\sqrt{\gamma}\Delta$ based on our reward distribution construction. Step (b) uses the fact that all base arms in a super arm have the same number of observations. Step (c) uses the lower bound constructed $1 - \sqrt{\frac{Q_A(t)+1}{Q_{A_1}(t)+1}} \geq \frac{1}{2}$. Note that for λ, the only randomness is $w \sim \mathcal{N}(0, 1)$ as all $Q_A(t)$ are determined by the history.

To construct an upper bound for λ above, we construct an optimization problem first using the constraint shown in (9). Recall (9) is $\sum_{A \in \Theta \setminus A_1} Q_A(t) \leq cT$. We construct the optimization problem with the objective function shown in the following equation:

$$\max_{x_1, x_2, \ldots, x_{K-1}} \Pr_{w \sim \mathcal{N}(0,1)}\left(w \leq 2\Delta\sqrt{x_a + 1}, \forall a \in [K-1]\right), \quad (14)$$

and constraints shown in (15)

$$x_a \geq 0, \forall a \in [K-1] \quad \text{and} \quad \textstyle\sum_{a=1}^{K-1} x_a \leq c \cdot T. \quad (15)$$

Note that the optimal solution to (14) is the same as the optimal solution to the following objective function (16):

$$\max_{x_1, x_2, \ldots, x_{K-1}} \Pr_{w \sim \mathcal{N}(0,1)}\left(w \leq \min_{a \in [K-1]} x_a\right). \quad (16)$$

It is not hard to verify that the objective function shown in (16) is maximized when $x_a = \frac{cT}{K-1} = \frac{c\sqrt{KT}}{(K-1)\Delta}$ for all $a \in [K-1]$. Therefore, $x_a = \frac{cT}{K-1} = \frac{c\sqrt{KT}}{(K-1)\Delta}$ for all $a \in [K-1]$ is also the optimal solution to (14) and the maximum value of the objective function shown in (14) is $\Pr_w\left(w \leq 2\Delta\sqrt{\frac{c\sqrt{KT}}{(K-1)\Delta} + 1}\right)$.

Now, we are ready to construct an upper bound for λ and have

$$\begin{aligned}
\lambda &= \Pr_w\left(w \leq 2\Delta\sqrt{Q_A(t)+1}, \forall A \in \Theta \setminus A_1 \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&\leq \max_{x_1, x_2, \ldots, x_{K-1}} \Pr_w\left(w \leq 2\Delta\sqrt{x_a+1}, \forall a \in [K-1] \mid \mathcal{F}_{t-1} = F_{t-1}\right) \\
&= \max_{x_1, x_2, \ldots, x_{K-1}} \Pr_w\left(w \leq 2\Delta\sqrt{x_a+1}, \forall a \in [K-1]\right) \\
&\overset{(a)}{\leq} \Pr_w\left(w \leq 2\Delta\sqrt{\frac{c\sqrt{KT}}{(K-1)\Delta} + 1}\right) \\
&\overset{(b)}{\leq} 1 - \frac{1}{8\sqrt{\pi}} \cdot e^{-\frac{7}{2} \cdot 4\Delta^2 \cdot \left(\frac{c\sqrt{KT}}{(K-1)\Delta} + 1\right)} \\
&\overset{(c)}{\leq} 1 - \frac{1}{8\sqrt{\pi}} \cdot e^{-\frac{7}{2} \cdot 4\Delta^2 \cdot \frac{2c\sqrt{KT}}{0.5K\Delta}} \\
&= 1 - \frac{1}{8\sqrt{\pi}} \cdot e^{-\frac{7}{2} \cdot 8\sqrt{\frac{K}{T}} \cdot \frac{1}{6} \cdot \frac{\sqrt{KT}}{0.5K}} \\
&= 1 - \frac{1}{8\sqrt{\pi}} e^{-\frac{28}{3}} = 1 - c_0,
\end{aligned} \quad (17)$$

where step (a) uses the fact that $\Pr_w\left(w \leq 2\Delta\sqrt{\frac{c\sqrt{KT}}{(K-1)\Delta} + 1}\right)$ is the maximum value of the objective function shown in (14). Step (b) uses the one-sided anti-concentration inequality shown in (7). Step (c) uses the fact that $K - 1 > 0.5K$, $\Delta = \sqrt{K/T}$, and when T is large enough, we have $\frac{c\sqrt{KT}}{(K-1)\Delta} \geq 1$.

By plugging the upper bound for λ into (13), we have $\Pr\left(\exists A \in \Theta \setminus A_1 : A_t = A \mid \mathcal{F}_{t-1} = F_{t-1}\right) \geq c_0$, which concludes the proof for the statement that with at least a constant probability c_0, a sub-optimal super arm is played in round t.

To complete the proof, we use the fact that the total regret from round $t = \alpha T$ to round T is at least $(1-\alpha)T \cdot c_0 \cdot m \cdot \sqrt{\gamma}\Delta = \Omega(Tm\Delta) = \Omega(Tm\sqrt{K/T}) = \Omega(Tm\sqrt{N/(mT)}) = \Omega(\sqrt{mNT})$, which is also a lower bound for the total regret from round 1 to round T. $\square$

Lemma 3. *We have* $\sum_{t=1}^{T}\sum_{a\in A_t}\sqrt{\frac{1}{n_{a,t}+1}} \leq 2\sqrt{mNT}$.

Proof of Lemma 3. Note that the LHS of the above inequality is a random variable. We provide an upper bound for this random variable.

Recall $n_{a,t} := \sum_{\tau=1}^{t-1}\mathbf{1}\left[a \in A_\tau\right]$ is the number of times that arm a has been played at the beginning of round t. Let $\tau_a(n)$ denote the round for arm a to be played for the n-th time, and thus $n_{a,\tau_a(n)} = n-1$.

$$\begin{aligned}
&\sum_{t=1}^{T}\sum_{a\in A_t}\sqrt{\frac{1}{n_{a,t}+1}} = \sum_{t=1}^{T}\sum_{a\in [N]}\sqrt{\frac{1}{n_{a,t}+1}}\mathbf{1}[a\in A_t] \\
&\overset{(a)}{=} \sum_{a\in[N]}\sum_{n=1}^{n_{a,T+1}}\sum_{t=\tau_a(n)}^{\tau_a(n+1)-1}\sqrt{\frac{1}{n_{a,t}+1}}\mathbf{1}[a\in A_t] \\
&\overset{(b)}{=} \sum_{a\in[N]}\sum_{n=1}^{n_{a,T+1}}\sqrt{\frac{1}{n}} \leq \sum_{a\in[N]}\int_0^{n_{a,T+1}}\sqrt{\frac{1}{n}}dn \\
&= 2\sum_{a\in[N]}\sqrt{n_{a,T+1}} \overset{(c)}{\leq} 2\sqrt{N\sum_{a\in[N]}n_{a,T+1}} \overset{(d)}{=} 2\sqrt{mNT},
\end{aligned} \tag{18}$$

where step (a) partitions all T rounds into multiple intervals based on the arrivals of observations from arm a. Step (b) uses the fact that $\sum_{t=\tau_a(n)}^{\tau_a(n+1)-1}\mathbf{1}[a\in A_t]\cdot\sqrt{\frac{1}{n_{a,t}+1}} = \sqrt{\frac{1}{n-1+1}} = \sqrt{\frac{1}{n}}$, because $n_{a,\tau_a(n)} = n-1$ and $\mathbf{1}[a \in A_t] = 0$ for all $t \in \{\tau_a(n)+1,\ldots,\tau_a(n+1)-1\}$. Step (c) uses the Cauchy-Schwarz inequality. Step (d) uses the fact that $\sum_{a\in[N]} n_{a,T+1} \leq mT$. $\square$

B. Proof of Lemma 2

Proof of Lemma 2. We give the details of the three steps as follows.

Step 1 proof. If $\mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t}\bar{r}_{a,t}\right] \leq 0$, the proof is trivial as the RHS in (4) is non-negative.

Recall $(\cdot)^+ := \max\{\cdot, 0\}$. For the case where $\alpha := \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t}\bar{r}_{a,t}\right] > 0$, we use Markov's inequality and have

$$\begin{aligned}
\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right]\right)^+\right] &\geq \alpha\cdot\Pr_{\Theta_t}\left(\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right]\right)^+ \geq \alpha\right) \\
&\geq \alpha\cdot\Pr_{\Theta_t}\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right] \geq \alpha\right),
\end{aligned} \tag{19}$$

which together with Lemma 5 gives

$$\begin{aligned}
\alpha &= \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t}\bar{r}_{a,t}\right] \\
&\leq \frac{\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right]\right)^+\right]}{\Pr_{\Theta_t}\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right] \geq \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t^*} r_a - \sum_{a\in A_t}\bar{r}_{a,t}\right]\right)} \\
&\leq \frac{\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right]\right)^+\right]}{\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}\bar{r}_{a,t} \geq \sum_{a\in A_t^*} r_a\right)} \\
&= 2\Phi(-\sqrt{4/\gamma})^{-1}\cdot\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right]\right)^+\right].
\end{aligned} \tag{20}$$

Step 2 proof. Since w_t and $\tilde{w}_t$ are i.i.d., we have $\mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right] = \mathbf{E}_{\Theta_t}\left[\max_{A\in\Theta_t}\sum_{a\in A}\bar{r}_{a,t}\right] = \mathbf{E}_{\Theta_t}\left[\max_{A\in\Theta_t}\sum_{a\in A}\tilde{r}_{a,t}\right] \geq \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\tilde{r}_{a,t}\mid A_t\right] = \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\tilde{r}_{a,t}\mid A_t, w_t\right]$. Then, we have

$$\begin{aligned}
&\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\bar{r}_{a,t}\right]\right)^+\right] \\
&\leq \mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\tilde{r}_{a,t}\mid A_t\right]\right)^+\right] \\
&= \mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \mathbf{E}_{\Theta_t}\left[\sum_{a\in A_t}\tilde{r}_{a,t}\mid A_t, w_t\right]\right)^+\right] \\
&= \mathbf{E}_{\Theta_t}\left[\left(\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \sum_{a\in A_t}\tilde{r}_{a,t}\right)\mid A_t, w_t\right]\right)^+\right] \\
&\leq \mathbf{E}_{\Theta_t}\left[\left|\mathbf{E}_{\Theta_t}\left[\left(\sum_{a\in A_t}\bar{r}_{a,t} - \sum_{a\in A_t}\tilde{r}_{a,t}\right)\mid A_t, w_t\right]\right|\right] \\
&\leq \mathbf{E}_{\Theta_t}\left[\mathbf{E}_{\Theta_t}\left[\left|\sum_{a\in A_t}\bar{r}_{a,t} - \sum_{a\in A_t}\tilde{r}_{a,t}\right|\mid A_t, w_t\right]\right] \\
&\leq \mathbf{E}_{\Theta_t}\left[\left|\sum_{a\in A_t}\bar{r}_{a,t} - \sum_{a\in A_t}\tilde{r}_{a,t}\right|\right].
\end{aligned} \tag{21}$$

Step 3 proof. By Hölder's inequality, we have that

$$\begin{aligned}
&\mathbf{E}\left[\sum_{t=1}^{T}\left|\sum_{a\in A_t}\bar{r}_{a,t}-\sum_{a\in A_t}\tilde{r}_{a,t}\right|\right]\\
&\leq \mathbf{E}\left[\sum_{t=1}^{T}|w_t-\tilde{w}_t|\sum_{a\in A_t}\sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}\right]\\
&\leq \mathbf{E}\left[\max_{t\in[T]}|w_t-\tilde{w}_t|\sum_{t=1}^{T}\sum_{a\in A_t}\sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}\right]\\
&\overset{(a)}{\leq} \mathbf{E}\left[\max_{t\in[T]}|w_t-\tilde{w}_t|\right]\cdot 2\sqrt{\gamma mNT\ln T}\\
&\overset{(b)}{\leq} 2\sqrt{\ln 2T}\cdot 2\sqrt{\gamma mNT\ln T}\leq 4\ln T\sqrt{2\gamma mNT},
\end{aligned}\tag{22}$$

where step (a) is due to Lemma 3, and step (b) is due to Fact 2 and $w_t-\tilde{w}_t$ is a Gaussian variable with variance 2. □

C. Proof of Lemma 4

Lemma 4. *In CL-SG, the regret of the deviation part is*

$$\mathbf{E}\left[\sum_{t=1}^{T}\left(\sum_{a\in A_t}\bar{r}_{a,t}-\sum_{a\in A_t}r_a\right)\mathbf{1}[\mathcal{E}_t]\right]\leq 4\ln T\sqrt{\gamma mNT}+2\sqrt{6mNT\ln T}.$$

Proof of Lemma 4. Recall that $\bar{r}_{a,t}=\hat{r}_{a,n_{a,t}}+w_t\sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}$. When $\mathcal{E}_t$ happens, we have that

$$\begin{aligned}
&\mathbf{E}\left[\sum_{t=1}^{T}\left(\sum_{a\in A_t}\bar{r}_{a,t}-\sum_{a\in A_t}r_a\right)\mathbf{1}[\mathcal{E}_t]\right]\\
&=\mathbf{E}\left[\sum_{t=1}^{T}\sum_{a\in A_t}\left(\hat{r}_{a,n_{a,t}}+w_t\sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}-\hat{r}_{a,n_{a,t}}+\sqrt{\frac{3\ln Nt}{n_{a,t}+1}}\right)\mathbf{1}[\mathcal{E}_t]\right]\\
&\leq\sqrt{\gamma\ln T}\mathbf{E}\left[\sum_{t=1}^{T}w_t\left(\sum_{a\in A_t}\sqrt{\frac{1}{n_{a,t}+1}}\right)\right]+\sqrt{6\ln T}\mathbf{E}\left[\sum_{t=1}^{T}\sum_{a\in A_t}\sqrt{\frac{1}{n_{a,t}+1}}\right],
\end{aligned}\tag{23}$$

where the last inequality is due to that $N\leq T$. Regarding the first item in RHS of (23), we can apply Hölder's inequality to have that

$$\begin{aligned}
&\mathbf{E}\left[\sum_{t=1}^{T}w_t\left(\sum_{a\in A_t}\sqrt{\frac{1}{n_{a,t}+1}}\right)\right]\\
&\leq\mathbf{E}\left[\max_{1\leq t\leq T}|w_t|\cdot\left|\sum_{t=1}^{T}\sum_{a\in A_t}\sqrt{\frac{1}{n_{a,t}+1}}\right|\right]\\
&\leq\mathbf{E}\left[\max_{1\leq t\leq T}|w_t|\cdot 2\sqrt{mNT}\right]\leq 4\sqrt{mNT\ln T},
\end{aligned}$$

where the second inequality is due to Lemma 3, and the last inequality is due to the maximal inequality (Fact 2) for Gaussian variables such that $\mathbf{E}\left[\max_{1\leq t\leq T}|w_t|\right]\leq\sqrt{2\ln 2T}\leq 2\sqrt{T}$.

Regarding the second term in RHS of (23), we can invoke Lemma 3 again to give a bound of $2\sqrt{6mNT\ln T}$.

□

D. Proof of Lemma 5

Lemma 5. *In each round $t>\max\{\sqrt{m},4\}$, given any Θ_t, we have that for CL-SG:*

$$\frac{1}{\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}\bar{r}_{a,t}\geq\sum_{a\in A_t^*}r_a\right)}\leq 2\Phi\left(-\sqrt{4/\gamma}\right)^{-1}.$$

Proof of Lemma 5. Given Θ_t, A_t^* is determined. Define $\mathcal{H}_t:=\left\{\forall a\in A_t^*:|r_a-\hat{r}_{a,n_{a,t}}|\leq\sqrt{\frac{4\ln t}{n_{a,t}+1}}\right\}$. We have

$$\begin{aligned}
\Pr_{\Theta_t}(\mathcal{H}_t)&\geq 1-\sum_{a\in A_t^*}\sum_{s_a=0}^{t-1}\Pr_{\Theta_t}\left(|r_a-\hat{r}_{a,s_a}|\geq\sqrt{\frac{4\ln t}{s_a+1}}\right)\\
&=1-\sum_{a\in A_t^*}\sum_{s_a=1}^{t-1}\Pr_{\Theta_t}\left(|r_a-\hat{r}_{a,s_a}|\geq\sqrt{\frac{4\ln t}{s_a+1}}\right)\\
&\geq 1-\sum_{a\in A_t^*}\sum_{s_a=1}^{t-1}\Pr_{\Theta_t}\left(|r_a-\hat{r}_{a,s_a}|\geq\sqrt{\frac{4\ln t}{2s_a}}\right)\\
&\geq 1-mt\cdot 2\cdot e^{-2\cdot s_a\cdot 4\ln t/(2s_a)}\\
&=1-\frac{2mt}{t^4}\\
&\geq 1-\frac{2}{t}\geq 0.5,
\end{aligned}\tag{24}$$

where the last two inequalities are due to that $t>\max\{\sqrt{m},4\}$. Then, we have

$$\begin{aligned}
&\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}\bar{r}_{a,t}\geq\sum_{a\in A_t^*}r_a\right)\geq\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}\bar{r}_{a,t}\geq\sum_{a\in A_t^*}r_a,\mathcal{H}_t\right)\\
&=\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}\bar{r}_{a,t}-\hat{r}_{a,n_{a,t}}\geq\sum_{a\in A_t^*}r_a-\hat{r}_{a,n_{a,t}},\mathcal{H}_t\right)\\
&=\Pr_{\Theta_t}(\mathcal{H}_t)\cdot\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}\bar{r}_{a,t}-\hat{r}_{a,n_{a,t}}\geq\sum_{a\in A_t^*}r_a-\hat{r}_{a,n_{a,t}}\mid\mathcal{H}_t\right)\\
&\overset{(a)}{\geq}0.5\cdot\Pr_{\Theta_t}\left(\sum_{a\in A_t^*}w_t\sqrt{\frac{\gamma\ln t}{n_{a,t}+1}}\geq\sum_{a\in A_t^*}\sqrt{\frac{4\ln t}{n_{a,t}+1}}\right)\\
&=0.5\cdot\Pr_{\Theta_t}\left(w_t\geq\sqrt{4/\gamma}\right)\\
&=0.5\cdot\Phi\left(-\sqrt{4/\gamma}\right),
\end{aligned}$$

where step (a) is due to (24) and the fact that event $\mathcal{E}_t$ is true. □

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